Bot 1 (pmus) Full Forensic Review & Fine-Tuning

HOST: AWS 52.16.177.132 (pmus.service @ 50ace40) | BANKROLL: ~$148 USDC | LOCAL VERIFICATION: 463 PASSED (0 FAILED)

Executive Verdict: Bot 1's core quadratic Liquidity Incentive Program (LIP) scoring and waterfill allocation math work as intended, generating **+$275.71** in verified non-trade rewards and rebates across 375 traded tokens and 1,593 fills. Net strategy PnL turned negative because market discovery admitted unseasoned (<48h) and short-expiry (<14d) jump events, sibling ladders concentrated up to 86.9% of capital on one underlying event, 0.01-tick thin top bids lacked a dollar front-shield, and underwater positions froze at breakeven asks while rotating active orders every 2 to 17 minutes. All seven root causes have been fixed and stress-tested locally inside `pmus/` without touching the live VPS.

1. Ground-Truth Wallet & Era Reconciliation

Operating Era	Tokens	Trade PnL	LIP + Rebates	Net PnL	Forensic Takeaway
Pre-LIP Testing (Oct 06-10)	237	-$190.24	$0.00	-$190.24	Legacy directional tests before LIP maker engine.
Single-Sided LIP (Oct 11-15)	34	-$144.25	+$62.80	-$81.45	1/3 single-sided penalty and naked directional fills.
Two-Sided Global (Oct 16-24, excl. PHP)	102	-$352.68	+$183.45	-$169.23	$22.39/day LIP on full days; dragged by early jump fills + $39.54 open MTM.
Two-Sided Global (Oct 16-24, incl. PHP)	104	-$411.83	+$183.45	-$228.38	Includes $59.15 locked in 2 halted PHP security-flaw positions.
Post-Wave-19 (>= Oct 21, excl. PHP)	32	-$91.48	+$81.90	**-$9.58**	Near breakeven once initial sports/weather/airdrop filters landed.

Non-Trade Income Verification: On-chain settled REWARD transfers total **+$234.2300** (matching official T+1 daily accruals to 4 decimals), plus **+$12.0198** accrued Oct 24 LIP, **+$19.4603** maker rebates, and **+$10.0000** taker rebates = **+$275.7101**.

2. What Bot 1 Does Right (Preserved Foundations)

1. Exact Quadratic LIP Formula (`global_rewards.py`): Faithfully computes Polymarket Global's quadratic order score, 1/3 single-sided divisor, and [0.10, 0.90] extreme-midpoint single-sided cutoff.
2. Concave Marginal Waterfilling (`condition_selector.py`): Steps across $36-$85 capital tiers by marginal reward per dollar rather than over-sizing one pool.
3. Cross-Book Synthetic Arb Shield (`book_guard.py`, `engine.py`): Enforces `price + opp_bid <= 0.99` both before entry and mid-tick.
4. Self-Order Stripping (`strip_orders_from_book`): Deducts own resting and grace-window orders before midpoint calculation so the bot never chases its own bids.

3. Seven Root Causes Identified & Remediated in pmus/

#	Subsystem	What Was Going Wrong	Implemented Fix in pmus/
1	`clob_venue.py` `models.py`	Admitted <48h unseasoned markets (-$156.59 incl. PHP), <14d expiries (-$91.20), and live delayed-matching games.	Surfaced native `/sampling-markets` fields with 0 extra HTTP calls; enforced 48h min age, 14d TTE (30d series futures), `seconds_delay == 0`, active flags, and 300s cache.
2	`condition_selector.py`	Allocated 83.8%-86.9% of bankroll into sibling AI Arena and Israeli Minister markets lacking `neg_risk_market_id`.	Dual correlation keys (`negrisk` + deadline/strike-normalized `question_stem`), 42% cap ($36 floor) when >=3 groups qualify, 1.3x admission hysteresis, and continuous risk-adjusted efficiency.

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3	`fill_risk.py` `keep_negatives.json`	43 mislabeled esports/IPO/box-office strings in `keep_negatives.json` blocked stronger filters, leaking 16 Oct 24 jump markets.	Surgically pruned 43 poisoned fixtures (preserving 334 clean cases) and expanded jump rules (AI leaderboards, valuations, Metascore, box-office gross, airspace, CVEs).
4	`global_rewards.py` `book_guard.py` `lip_quoter.py`	No $25 step-back on 1-level/0.03-spread thin top bids, no ask-overhang guard, and no live [0.22, 0.78] pair bounds (Markus Ruck @ 0.218).	Added multi-level `_has_front_shield` (>=$25) with 0.03-spread half-cent step-back, >12.5x ask-overhang reject (>=$500 ask vs `near_bid_usd`), and live [0.22, 0.78] pair bounds.
5	`lip_quoter.py` `engine.py`	Underwater bags posted dead breakeven asks forever; manual below-avg sells were canceled; rotated pairs risked unhedged stop-sells.	4-state `UnwindRegime` (`HEDGED_PAIR_HOLD` persisted via `known_sibling_by_slug`, `IN_SPREAD_SCRATCH`, `LIQUID_STOP_UNWIND` down to 0.55x avg when ask >= stop_floor, `COLLAPSED_STUB_HOLD`) + manual sell protection.
6	`engine.py`	0.92 utilization threshold rotated targets 13 times in 3h41m, breaking 60s continuous LIP scoring cycles.	Lowered `CAPITAL_UTILIZATION_RATIO` to 0.75 (30 ticks, 180s cooldown) and added asymmetric `reconcile_quotes` tolerance (<=+20% upsize kept; immediate downsize cancel).
7	`ledger.py`, `shadow.py` `config.py`, `report.py`	Overnight cost basis reported $0 loss after UTC midnight; `get_lip_stats(since_ms>0)` undercounted seconds 10x-30x.	Seeded pre-window cost basis ($16.90 tracked), added UTC `daily_loss_baseline_date` + dynamic daily stop, persisted `dt_sec` in `lip_snapshots`, and stripped mirrored cross-side asks.

4. Verification Scorecard (Swarm + Arena + Loop + Multi-Model Interrogate)

Verification Gate	Baseline (50ace40)	Tuned pmus/ Worktree
Unit & Regression Suite (`pytest pmus/tests -q`)	`442 passed`	**`463 passed in 7.53s (0 failed)`**
Historical 54 Two-Sided Losers Blocked	`24 / 54 (53.82% USD)`	**`51 / 54 (96.31% USD)`**
Oct 24 Leaked Jump Targets + Live Pool Substitutions Blocked	`0 / 16 (0.0%)`	**`16 / 16 (100.0%) + 3 Live Rank Leaks`**
Clean Institutional Negative Fixtures (`keep_negatives.json`)	`377 (43 mislabeled)`	**`334 / 334 valid cases green`**
Live 3,000-Book Allocation ($148.07 Available Budget)	Single-theme jump concentration	**`$144.25 (97.42%) across 3 uncorrelated pairs`**
Live 3,000-Book Expected LIP Yield & Pair Spread Buffer	Churned every 2-17 min	**`$15.94 / day (5c to 8c merge margin per pair)`**

5. Operator Inventory Guidance & Safe Deployment Checklist

1. Existing Open Bags on Wallet: Upon restart, `Alibaba` (-$16.58 MTM, Oct 31 expiry) and `Kimi K3.1 Pro` (-$9.40 MTM) sit on liquid books above `0.55 * avg_price` and will actively unwind via `LIQUID_STOP_UNWIND` unless you place manual limit `SELL` orders on Polymarket first (which the engine adopts and protects). `Markus Ruck` ($0.01 stub) enters `COLLAPSED_STUB_HOLD` and will not dump. Halted PHP shares ($59.15) remain untouched awaiting UMA settlement.
2. Clean `/etc/pmus.env` on VPS: Remove stale `PMUS_DAILY_LOSS_BASELINE=72.62` so the dynamic daily loss stop resets cleanly at UTC midnight.
3. Safe Deploy Sequence (Pending Your Explicit Approval): Stage only `git add pmus/` (leaving root legacy scripts unstaged). On `52.16.177.132`, run `sudo systemctl stop pmus.service` **before** pulling and starting so `ALTER TABLE lip_snapshots ADD COLUMN dt_sec` runs cleanly without lock contention on the 2.6 GB SQLite WAL database.